3W Well Data Viewer

A desktop GUI to inspect the real instances of the 3W dataset

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Overlapping samples

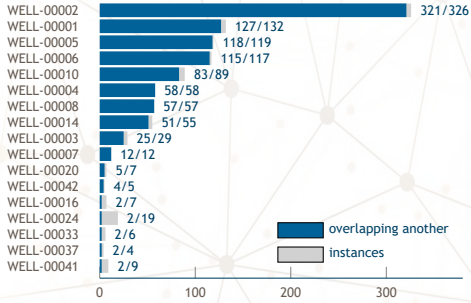
Timelines page ► How are the instances of one well distributed in time?

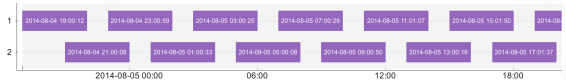
17
of 40 wells have overlapping instances

986
of 1,119 instances overlap another of their well

0
label collisions in the 912 overlapping pairs

Instances of the wells with overlaps (the other 23 have none)





Instances overlapping in WELL-00004.

Worries:

- Overlapping samples may land in **train and test**, causing leakage.
- Duplicated stretches pull **clusters** tighter, inflating scores.

Thankfully, 3W 2.0.0 has no label collisions, making joins principled and avoiding leakage.
Clustering must still be carefully calculated!

What a joined history is labeled as

Timelines page ▶ How are the instances of one well distributed in time?

12.3 % ▷ 2.9 %

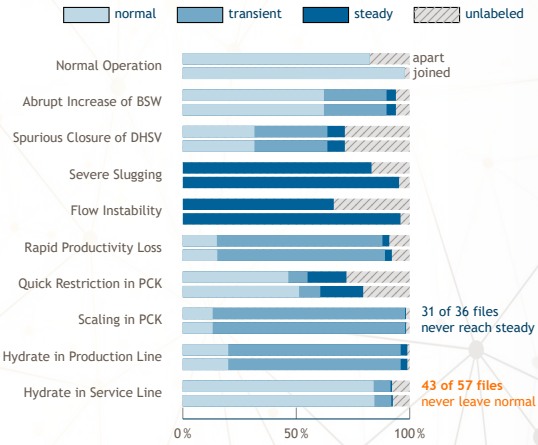
of unlabels after join

After the join, most unlabeled stretches appear trivial and might be recoverable.

Unlabeled stretch position	*n*	Duration	Share
head of a recording	208	1 h each	86.5 %
seam between equal labels	882	median 76 s, max 56 min	11.5 %
between two labels	16	median 6 min, max 1 h	2.0 %

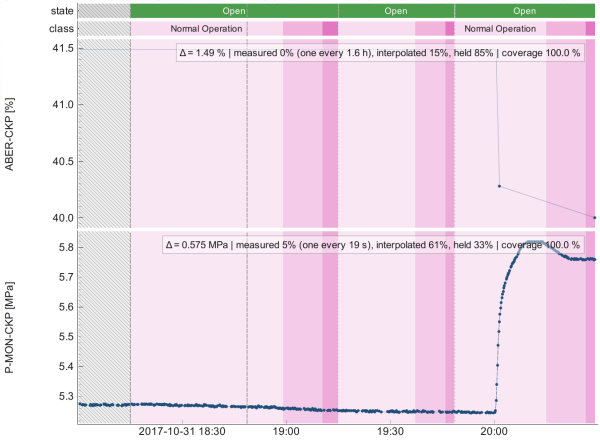
Appendix

Each fault folder's recorded time, by label, its files apart and joined



Histories whose labels do not add up

Timelines page ► How are the instances of one well distributed in time?



WELL-00004's joined Quick Restriction in PCK instances.

- Joined histories go back to normal **13 times** in 7 wells, **every time at a file boundary**.
- **WELL-00004**, left: three quick restrictions in 75 min, each back to normal 5 to 15 s after its steady state.
- **WELL-00006**: Abrupt Increase of BSW, steady, then normal 30 min later in a Scaling in PCK file.
- The others: Flow Instability (WELL-00001, 00004), Rapid Productivity Loss (00016, 00020), Scaling in PCK (00024), Hydrate in Service Line (00042).

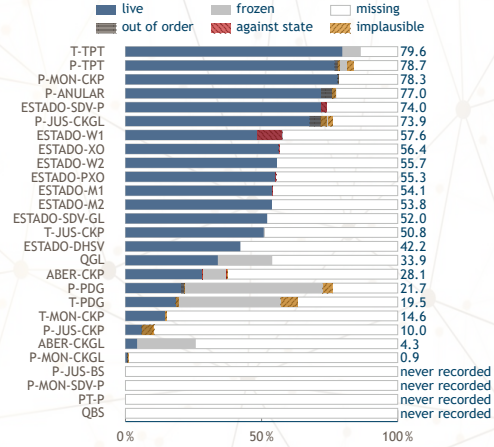
▸ Appendix

What each variable recorded

Availability page ► How well do sensors and states cover the dataset?

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See the detailed spread in the Appendix



PL check P-PDG → P-TPT → P-MON-CKP → *PCK open* → P-JUS-CKP

SL check P-MON-CKGL → *CKGL open* → P-JUS-CKGL ← *M2, W2 open* ← P-ANULAR

Valve checks

Open	M1, W1, SDV-P, PCK open; PXO, XO closed
Restart	M1, W1, SDV-P, PCK open
Shut-In, Closed With Diesel or Gas	one of M1, W1, SDV-P, PCK closed
Flushing Diesel or Gas	M1 or W1 closed; PXO or XO open
Depressurization	SDV-P, PCK open; M1 or W1 closed; PXO, XO closed

Plausible ranges

Pressure	Temperature	Valve opening	Flow rate	Valve state
0 to 100 MPa	−50 to 250 °C	0 to 100 %	0 to 10 m^3/s	$<10^8$

PETROBRAS

Thank you!



github.com/Thiago-TP/3w-well-data-viewer

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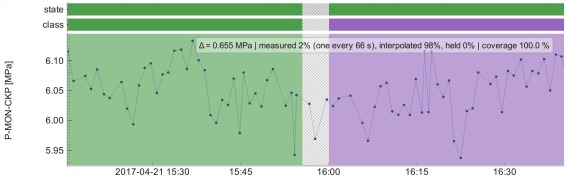
Appendix: label changes across an unlabeled stretch

Timelines page ► How are the instances of one well distributed in time?

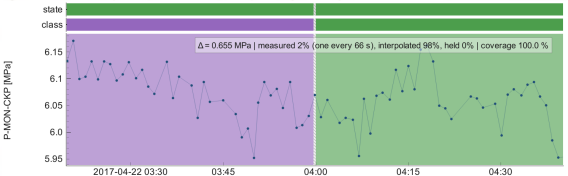
The 16 unlabeled stretches of a joined history between two different labels.

	Well	Unlabeled from	Label before	Label after	For	Shown on
1	WELL-00001	2017-04-21 15:55	normal	Flow Instability (steady)	4 min	P-MON-CKP
2	WELL-00001	2017-04-22 03:59	Flow Instability (steady)	normal	25 s	P-MON-CKP
3	WELL-00001	2017-06-26 09:59	Flow Instability (steady)	normal	3 min	P-MON-CKP
4	WELL-00002	2013-12-12 08:59	normal	Flow Instability (steady)	41 s	P-TPT
5	WELL-00002	2013-12-14 22:00	normal	Flow Instability (steady)	9 s	P-ANULAR
6	WELL-00004	2014-08-06 22:00	Flow Instability (steady)	normal	32 s	P-MON-CKP
7	WELL-00004	2017-10-31 19:14	Quick Restriction in PCK (steady)	normal	5 s	P-MON-CKP
8	WELL-00004	2017-10-31 19:48	Quick Restriction in PCK (steady)	normal	15 s	P-MON-CKP
9	WELL-00006	2018-06-18 11:00	Abrupt Increase of BSW (steady)	normal	30 min	P-MON-CKP
10	WELL-00016	2018-04-26 14:44	Rapid Productivity Loss (transient)	normal	7 min	P-MON-CKP
11	WELL-00020	2014-03-18 01:30	Rapid Productivity Loss (steady)	normal	29 min	P-MON-CKP
12	WELL-00020	2014-03-18 13:57	Rapid Productivity Loss (transient)	normal	53 min	P-MON-CKP
13	WELL-00020	2014-03-19 11:00	Rapid Productivity Loss (transient)	normal	1 h	P-MON-CKP
14	WELL-00024	2016-09-11 07:00	Scaling in PCK (transient)	normal	40 min	P-MON-CKP
15	WELL-00042	2014-12-18 00:53	Hydrate in Service Line (transient)	normal	48 min	P-MON-CKP
16	WELL-00042	2014-12-22 03:37	Hydrate in Service Line (transient)	normal	8 min	P-MON-CKP

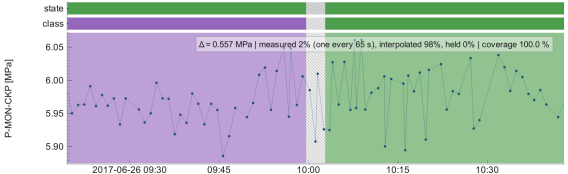
Appendix: label changes 1 to 4 (of 16)



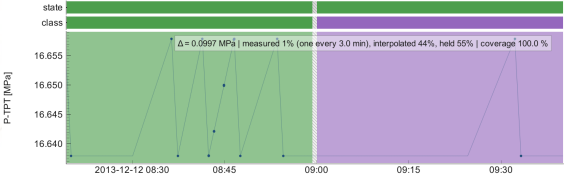
1 WELL-00001, 2017-04-21 15:55: normal → 4 min unlabeled → Flow Instability (steady)



2 WELL-00001, 2017-04-22 03:59: Flow Instability (steady) → 25 s unlabeled → normal

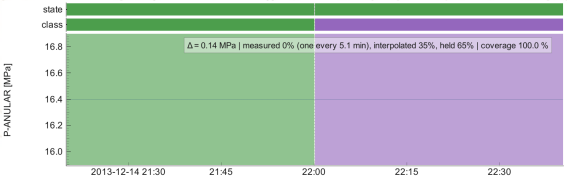


3 WELL-00001, 2017-06-26 09:59: Flow Instability (steady) → 3 min unlabeled → normal

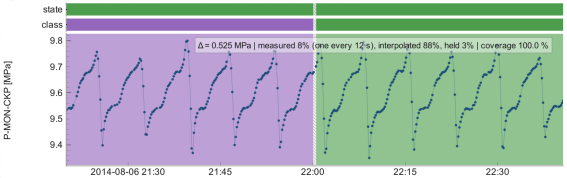


4 WELL-00002, 2013-12-12 08:59: normal → 41 s unlabeled → Flow Instability (steady)

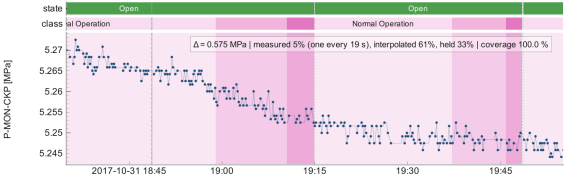
Appendix: label changes 5 to 8 (of 16)



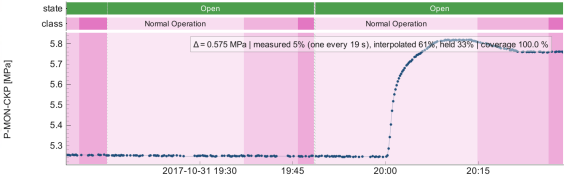
5 WELL-00002, 2013-12-14 22:00: normal → 9 s unlabeled → Flow Instability (steady)



6 WELL-00004, 2014-08-06 22:00: Flow Instability (steady) → 32 s unlabeled → normal

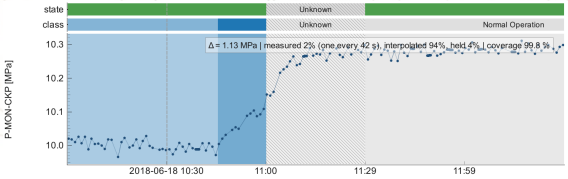


7 WELL-00004, 2017-10-31 19:14: Quick Restriction in PCK (steady) → 5 s unlabeled → normal

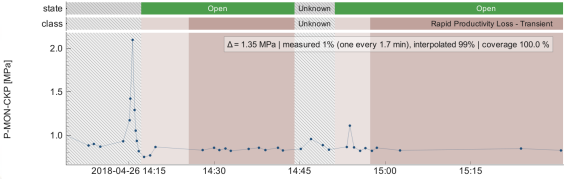


8 WELL-00004, 2017-10-31 19:48: Quick Restriction in PCK (steady) → 15 s unlabeled → normal

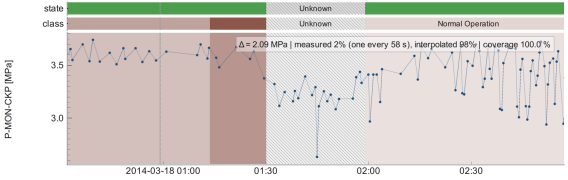
Appendix: label changes 9 to 12 (of 16)



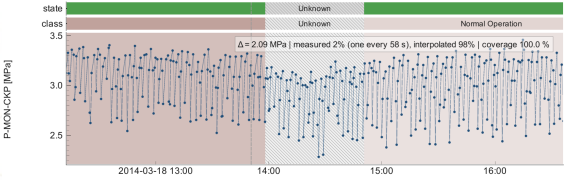
9 WELL-00006, 2018-06-18 11:00: Abrupt Increase of BSW (steady) → 30 min unlabeled → normal



10 WELL-00016, 2018-04-26 14:44: Rapid Productivity Loss (transient) → 7 min unlabeled → normal

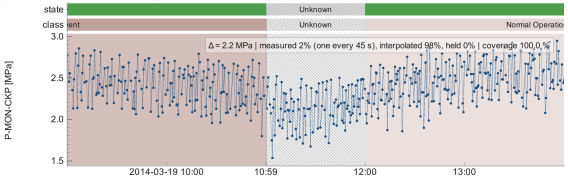


11 WELL-00020, 2014-03-18 01:30: Rapid Productivity Loss (steady) → 29 min unlabeled → normal

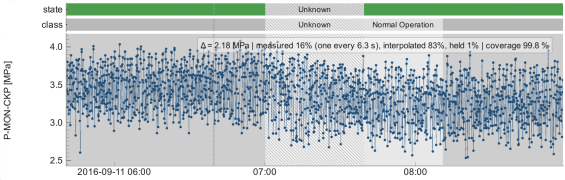


12 WELL-00020, 2014-03-18 13:57: Rapid Productivity Loss (transient) → 53 min unlabeled → normal

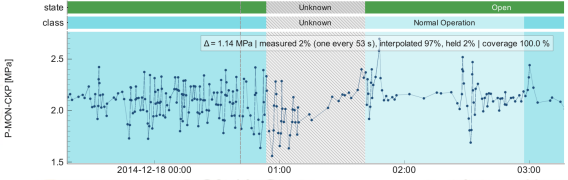
Appendix: label changes 13 to 16 (of 16)



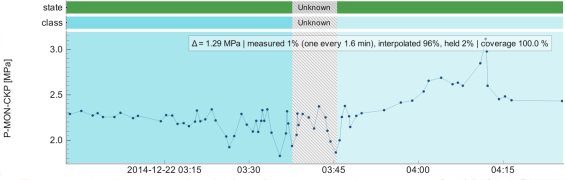
13 WELL-00020, 2014-03-19 11:00: Rapid Productivity Loss (transient) → 1 h unlabeled → normal



14 WELL-00024, 2016-09-11 07:00: Scaling in PCK (transient) → 40 min unlabeled → normal



15 WELL-00042, 2014-12-18 00:53: Hydrate in Service Line (transient) → 48 min unlabeled → normal



16 WELL-00042, 2014-12-22 03:37: Hydrate in Service Line (transient) → 8 min unlabeled → normal

Appendix: what each variable recorded

Availability page ► How well do sensors and states cover the dataset?

Variable	Live			Frozen			Implausible			Out of order			Against state			Missing			Farthest out
	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	
T-TPT	26,178,605	79.6	1,006	2,236,395	6.8	29	0	0.0	0	-	-	-	-	-	-	4,457,250	13.6	84	
P-TPT	25,878,401	78.7	922	1,612,066	4.9	38	1,004,849	3.1	10	361,705	1.1	29	-	-	-	5,381,783	16.4	159	2942 MPa
P-MON-CKP	25,749,471	78.3	720	42,928	0.1	2	12,489	0.0	5	104,739	0.3	6	-	-	-	7,079,851	21.5	397	−0.237 MPa
P-ANULAR	25,320,751	77.0	914	139,764	0.4	7	387,650	1.2	3	1,296,207	3.9	6	-	-	-	7,411,735	22.5	198	−7.658 MPa
ESTADO-SDV-P	24,316,107	74.0	661	0	0.0	0	0	0.0	0	-	-	-	653,430	2.0	23	8,556,143	26.0	458	
P-JUS-CKGL	24,307,254	73.9	656	743,125	2.3	33	1,299,132	4.0	85	1,298,303	4.0	7	-	-	-	7,821,871	23.8	430	−0.4977 MPa
ESTADO-W1	18,938,036	57.6	558	0	0.0	0	0	0.0	0	-	-	-	3,086,682	9.4	14	13,934,214	42.4	561	
ESTADO-XO	18,538,730	56.4	533	0	0.0	0	0	0.0	0	-	-	-	3,351	0.0	1	14,333,520	43.6	586	
ESTADO-W2	18,293,462	55.7	533	0	0.0	0	0	0.0	0	-	-	-	-	-	-	14,578,788	44.4	586	
ESTADO-PXO	18,176,203	55.3	532	0	0.0	0	0	0.0	0	-	-	-	57,977	0.2	3	14,696,047	44.7	587	
ESTADO-M1	17,773,205	54.1	501	0	0.0	0	0	0.0	0	-	-	-	8,889	0.0	3	15,099,045	45.9	618	
ESTADO-M2	17,688,282	53.8	504	0	0.0	0	0	0.0	0	-	-	-	-	-	-	15,183,968	46.2	615	
ESTADO-SDV-GL	17,105,271	52.0	480	0	0.0	0	0	0.0	0	-	-	-	-	-	-	15,766,979	48.0	639	
T-JUS-CKP	16,689,089	50.8	554	150,261	0.5	8	0	0.0	0	-	-	-	-	-	-	16,032,900	48.8	557	
ESTADO-DHSV	13,865,397	42.2	470	0	0.0	0	0	0.0	0	-	-	-	-	-	-	19,006,853	57.8	649	
QGL	11,150,903	33.9	147	6,569,202	20.0	298	0	0.0	0	-	-	-	-	-	-	15,152,145	46.1	674	
ABER-CKP	9,248,257	28.1	272	3,078,201	9.4	27	161,872	0.5	2	-	-	-	11,955	0.0	4	20,545,792	62.5	820	−99.99 %
P-PDG	7,119,658	21.7	174	17,836,376	54.3	653	1,200,682	3.7	33	256,966	0.8	24	-	-	-	7,916,216	24.1	292	-1.2×10^{36} MPa
T-PDG	6,426,251	19.5	142	14,385,021	43.8	421	2,445,015	7.4	36	-	-	-	-	-	-	12,060,978	36.7	556	-1.7×10^{38} °C
T-MON-CKP	4,789,008	14.6	14	161,538	0.5	1	161,538	0.5	1	-	-	-	-	-	-	27,921,704	84.9	1,104	−99.99 °C
P-JUS-CKP	3,287,363	10.0	10	161,538	0.5	1	1,506,941	4.6	7	0	0.0	0	-	-	-	29,423,349	89.5	1,108	1559 MPa
ABER-CKGL	1,403,126	4.3	48	7,088,387	21.6	238	0	0.0	0	-	-	-	-	-	-	24,380,737	74.2	833	
P-MON-CKGL	308,726	0.9	2	0	0.0	0	6,401	0.0	1	2,096	0.0	1	-	-	-	32,563,524	99.1	1,117	−0.094 MPa
P-JUS-BS	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
P-MON-SDV-P	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
PT-P	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
QBS	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	